

Daily Portfolio Digest

A simple setup guide for Google Sheets + Google Apps Script

What this builds: an automated daily email that reads your current portfolio from a Google Sheet, finds company and macro-market news, uses Gemini to identify likely catalysts and write concise commentary, then emails you a formatted portfolio digest each morning.

1. What you need

You need a Google account, a Google Sheet containing your holdings, a Finnhub API key, a Google Gemini API key, and an email address where you want the digest delivered. The portfolio itself stays in the spreadsheet - you do not need to hard-code your holdings into the script.

2. Create the Google Sheet

Create a new Google Sheet and use these five columns, starting in row 1:

Column	Heading	Purpose
A	Ticker	Stock / ETF ticker
B	Price	Current market price
C	Change %	Daily percentage change
D	Headlines	Optional existing headline field
E	Share Name	Full company / fund name

Important: the script reads every populated holding from columns A-E. If you later add or remove a stock from the sheet, the digest adapts automatically. Column D is treated as Headlines - not quantity.

Populate the Price and Change % columns using your preferred Google Sheets market-data formulas or data source. The digest code expects numeric values in B and C; the exact formula depends on the market/instrument you use.

3. Get your API keys

Finnhub: create an account with Finnhub and generate an API key. This is used primarily for company news. The script also uses Google News RSS as a fallback/supplement because free Finnhub access can be rate-limited.

Gemini: create a Gemini API key through Google's AI developer tooling. Gemini is used to turn the retrieved news into specific catalyst descriptions and analytical market commentary.

Do not share your API keys. If you post the script publicly or send it to friends, replace your keys and email address with placeholders first.

4. Add the Apps Script

Open your portfolio Sheet, then go to **Extensions > Apps Script**. Delete the starter code and paste in the full Daily Portfolio Digest script.

At the top of the script, replace the placeholders:

```
const FINNHUB_API_KEY = "YOUR_FINNHUB_KEY_HERE";
const GEMINI_API_KEY = "YOUR_GEMINI_KEY_HERE";
const EMAIL = "YOUR_EMAIL@gmail.com";
```

Save the project. You can give it a name such as **Daily Portfolio Digest**.

5. Run it manually the first time

In Apps Script, select **dailySummary** from the function menu and click **Run**. Google will ask you to authorise access to the spreadsheet, external web requests and email sending. Approve the permissions for your own script.

When the run completes successfully, check your inbox. You should receive a formatted email containing an opening market overview, top movers, sector commentary, notable declines, ETFs/funds and related news sources.

6. Turn on the daily email

Once the manual test works, select **createDailyTrigger** and run it once. This removes any previous **dailySummary** trigger and creates a new daily trigger for roughly 08:07.

Apps Script time-based triggers are not guaranteed to execute at the exact second, so treat 08:07 as an approximate morning delivery time.

7. How the digest works

	Stage	What happens
1	Reads holdings	Pulls tickers, prices, daily changes and names directly from your Sheet.
2	Finds movers	Separates individual stocks from funds and identifies the largest gains and declines.
3	Finds company news	Combines FintHub and Google News, including catalyst searches for upgrades, downgrades, earnings, deals and approvals.
4	Finds macro news	Searches broad markets, rates, inflation, oil, geopolitics, trade, semiconductors, AI and UK-market stories.
5	Uses Gemini	Selects plausible stock-specific catalysts and writes market/portfolio commentary without inventing unsupported explanations.
6	Builds the email	Formats prices, changes, source links and commentary into an HTML digest and sends it automatically.

8. Currency handling

The supplied script attempts to display ordinary non-UK equities in USD, UK-listed instruments in GBP and relevant European companies in EUR. This is heuristic rather than exchange-aware, so unusual listings may need an override.

```
const CURRENCY_OVERRIDES = {  
  "XYZ": { currency: "EUR", divisor: 1 },  
  "ABC": { currency: "GBP", divisor: 100 }  
};
```

You normally leave this object empty. It does **not** define your portfolio; it only corrects display currency for exceptional instruments.

9. Common troubleshooting

Finnhub says 429 / API limit reached

This means the free API quota has been exhausted. The script is designed to stop further Finnhub requests for that run and rely more heavily on Google News. Avoid repeatedly running the script while testing, or wait for the quota to reset.

Finnhub returns HTML / <!DOCTYPE> instead of JSON

This can happen with unsupported or flaky endpoints. The current version checks for HTML before attempting JSON parsing. It does not rely on Finnhub's news-search endpoint.

Utilities.sleep says the parameter is null

Use the current version's `sleepMs_()` wrapper. It converts the delay to a number and simply returns if the value is missing or invalid, preventing `Utilities.sleep(null)`.

A mover still shows N/A

N/A is intentional when the retrieved stories do not provide a credible catalyst for the direction of the move. This is preferable to attaching an unrelated article. The script searches both general company news and catalyst terms such as analyst upgrades/downgrades, price targets, earnings, guidance, contracts, acquisitions and approvals.

A UK price looks 100x too high or too low

Some UK market feeds quote prices in pence rather than pounds. Add a currency override with divisor 100 if the automatic heuristic gets a particular instrument wrong.

The email doesn't arrive

Check Apps Script's **Executions** page for errors, confirm the EMAIL constant, verify that the first-run permissions were approved, and check spam/junk folders.

10. Updating your portfolio later

Just edit the Google Sheet. Add a new row for a new holding or remove a row for something you sold. You do not need to edit the Apps Script simply because your portfolio changed.

11. Before sharing your copy

Remove your Finnhub key, Gemini key and personal email address. Each person should create their own Sheet and their own API credentials. API quotas and availability can change, particularly on free tiers.

Quick checklist

1. Create the five-column Google Sheet.
2. Add your holdings and working Price / Change % data.
3. Get a Finnhub API key.
4. Get a Gemini API key.
5. Open Extensions > Apps Script.
6. Paste the full digest code.
7. Insert your own keys and email address.
8. Run dailySummary once and approve permissions.
9. Check the test email.
10. Run createDailyTrigger once.
11. From then on, maintain the portfolio in the Sheet.

Note: This project is a personal automation template, not investment advice. News feeds and AI-generated summaries can miss context or identify an imperfect catalyst, so source links are included for verification.